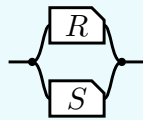


Relation Algebra

§4. $\cap$ is a commutative idempotent monoid on every hom-set

(4a)

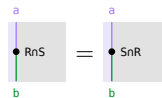
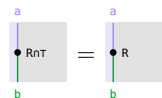
The **meet** of $R, S : a \rightarrow b$, the paper's **convolution**, is $R \cap S := \langle (R \otimes S) \rangle$ — copy the input, run R and S on the two copies, merge the results — so what comes out is what both of them do.



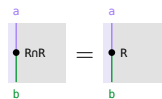
transcribed: a definition has no statement to export

On every hom-set it is associative, commutative and idempotent, with unit the maximal arrow $T = \text{---} \circ \text{---}$ (the paper's Lemma 4.11).

(4b)



unit: one half of T per end — the merge's unit law absorbs the $\text{---} \circ$, the copy's counit law the $\text{---} \circ$ **commutative:** σ crosses $R \otimes S$ by naturality and is absorbed by cocommutativity and commutativity



associative: coassociativity and associativity; $\otimes$ re-brackets for nothing, **idempotent:** the one that is not bookkeeping — the lax copy law is the whole of it, worked in allegory2

So $\leq$ is the order this monoid induces. $R \cap S \leq R$ comes from the unit, and idempotency turns anything under both S and T into something under $S \cap T$, since $R = R \cap R \leq S \cap T$.

And one law relating $\cap$ to composition, which is **not** an equation:

(4c)

semi-distributivity, and what supplies it	picture
$R \cap (S \cap T) \leq R \cap S \cap T$ — the lax copy law. Equality exactly when R is single valued: the Maps section's $F(R \cap S) = F R \cap F S$.	